

Practice Questions Developmental Genetics with Answers

1. The parents of a new-born child come to a physician because their child was born with a congenital malformation. The child's condition is caused by a genetically controlled event which led to an abnormality in the cellular organization of the long bones. Specifically, this mutation caused a premature ossification of the growth plate during fetal development. What best characterizes the abnormality in this infant?
 - A. Disruption
 - B. Syndrome
 - C. Association
 - D. Dysplasia
 - E. Deformation
2. A 33-year-old woman has given birth to a child with holoprosencephaly (HPE). During a discussion regarding prognosis the physician notes that the mother has a single upper incisor tooth. Which of the following genes most likely has a mutation which leads to the disorder in this child?
 - A. *FGFR3*
 - B. *SHH*
 - C. *CFTR*
 - D. *DMD*
 - E. *RB*
 - F. *TP53*
 - G. *HOXA13*
3. A researcher studying human development notes that a family containing many members who have an extra set of ribs, also have a higher risk of cancer. Genetic studies show that the affected members of this family all carry a pathogenic variant in a gene that contains a homeodomain. Which of the following best describes the function of the gene that causes the disorder in the affected individuals of this family?
 - A. A regulator of gene expression
 - B. A protease that cleaves a zymogen
 - C. A receptor protein kinase
 - D. A structural protein in skeletal muscle
 - E. A membrane bound chloride channel
4. A researcher studying human development notes that proper pattern formation relies on appropriate activity of the HOX genes. Which of the following best describes how these regulatory genes are arranged in the human genome?
 - A. Localized to the mitochondria
 - B. Dispersed in the genome
 - C. One on each chromosome
 - D. All on the sex chromosome
 - E. Arranged in clusters

5. A 32-year-old woman brings her new-born baby to the physician for follow-up because of dwarfism. Genetic testing shows a mutation that causes a Gly380Arg change in the FGFR3 gene. Which of the following best describes the likely effect of this variant on the encoded protein?

- A. Deletion of the active site
- B. Retention in the endoplasmic reticulum
- C. Degradation by the proteasome
- D. Persistent kinase activity
- E. Inappropriate interaction with RAS

6. A 14-year-old boy is brought to the physician by his parents for a two year history of decreased hearing in both ears. Physical examination shows a distinct white forelock of hair in the midline with brilliant bilateral blue iris, wide spacing between the inner corners of the eyes (telecanthus or dystopia canthorum) and patches of skin depigmentation. A blood sample is drawn and a targeted panel of HOX genes are selected for exome sequencing. A heterozygous mutation is found in the PAX3. Which of the following best describes the normal function of this gene?

- B. Connective tissue component
- C. Receptor tyrosine kinase
- D. Soluble morphogen
- E. Transcription factor
- F. RAS GTP regulator

Answers and explanations (developmental genetics)

1. Answer D. Explanation: Child has achondroplasia – a dysplasia - the FGFR3 mutation leads to an abnormal tissue structure (bone) caused by abnormal tissue (or cellular) organization. Disruption – destruction of a tissue or organ, after it has formed – usually as a result of external forces (amniotic bands) – often not genetic. Deformation – mechanical distortion by a physical force occurring while the tissue is forming.

2. Answer B. Explanation: There are many potential causes of holoprosencephaly, which is failure, or incomplete separation of the forebrain to form 2 hemispheres. Haploinsufficiency of SHH (sonic hedgehog) is a monogenic cause of holoprosencephaly. Note that there are many potential genetic contributors to this condition, and many unexplained causes, as well. Excessive signaling through FGFR3 leads to achondroplasia. Biallelic loss of function of CFTR causes cystic fibrosis. Mutation of the X-linked DMD gene causes Duchenne, or Becker muscular dystrophy. RB and TP53 are classic tumor suppressor genes. The HOX genes are found as clusters and they encode transcription factors (regulators of gene expression) which are required for pattern formation.

3. Answer A. Explanation: The homeodomain is found in proteins that bind to DNA and act as transcriptional regulators which is often an important aspect of gene expression (DNA – RNA – Protein). A zymogen is an inactive precursor protein, that upon cleavage becomes irreversibly activated. Examples include trypsinogen, which is the inactive, precursor form of trypsin. Once formed in the pancreas, trypsinogen must get to the duodenum where it can be activated. This mechanism helps to ensure that newly formed trypsinogen does not cause degradation of the pancreas. Another example of zymogens is

the caspase cascade that controls apoptosis. An example of a receptor protein kinase is FGFR3; mutagenic activation (loss of regulated activity) of FGFR3 causes achondroplasia. There are many structural proteins in skeletal muscle (of course). The dystrophin protein (encoded by DMD), links the actin cytoskeleton to the extracellular matrix to provide structural integrity to the muscle fiber. CFTR is a membrane bound chloride channel which must function to prevent cystic fibrosis.

4. Answer E. Explanation: The HOX genes are arranged in 4 clusters that are sequentially, and temporally expressed during fetal development. This is a unique and special event that occurs during development. Most genes that must be expressed during development are not found next to each other, or arranged in such clusters. One other exception is the beta-globin cluster required for hemoglobin formation.

5. Answer D. Explanation: Achondroplasia occurs because of premature ossification of the growth plate of the long bones due to persistent, unregulated, (mutagenically activated) FGFR3 activity. Note that this is a gain of function, which usually implies that a specific variant (mutation) occurs to cause this increased or “new” activity. Note that a loss of function of a gene can occur by millions of different mechanisms.

6. Answer E. Explanation: PAX3 is a transcription factor that belongs to the paired box (PAX) family of proteins. The gene located on chromosome 2q35 and the protein plays a critical role in the formation of tissues and organs during embryonic development. It regulates the expression of genes that signal the neural crest cells to form specialized tissues and melanocytes which contribute to hair, eye, and skin color as well as cells of the inner ear. The PAX3 transcription factor is a homodimer but can form heterodimers with other members of the PAX family members. Mutations in PAX3 can be inherited in an autosomal dominant fashion and is associated with Waardenburg syndrome types I and II where defects in melanocyte development lead to pigmentation anomalies and congenital deafness.